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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

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Reviewer

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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

This thesis addresses the heavy disease burden of lung cancer (China's leading cause of cancer incidence and mortality) and rapidly rising thyroid cancer (3rd most common in China, 466,000 new cases in 2022) by tackling key clinical bottlenecks in chest CT utilization—single examination targets leaving thyroid as a diagnostic blind spot, insufficient quantitative data mining, single-modality limitations, and lack of dynamic pulmonary nodule assessment—through three interconnected research directions following the logical framework of spatial expansion, modal deepening, and temporal extension: developing the DVT model using Transformer architecture and secondary transfer learning to achieve 96% accuracy and 0.97 AUROC for thyroid cancer screening from routine chest CT, exploring multimodal CT-CLIP vision-language models to integrate imaging and pathological text for lung cancer subtype classification, and creating the CL-IJLO model with contrastive learning-based 80% feature compression and combined loss functions to predict pulmonary nodule malignancy trajectories with 87% accuracy, 0.90 AUC, and 97% sensitivity while improving training efficiency by 2.1×, collectively providing a comprehensive intelligent diagnostic

paradigm from methodological innovation to multi-center clinical validation with significant theoretical and clinical value for advancing early cancer detection.

Knowledge Base and Literature Review.

This literature review demonstrates exceptional quality through its well-structured, logically progressive organization that systematically covers the complete paradigm shift from traditional radiomics to advanced deep learning architectures including CNNs and Transformers, while comprehensively addressing both thyroid and lung cancer AI diagnostic applications; its greatest strength lies in its precise identification of three critical research gaps—the underutilization of incidental thyroid regions on routine chest CT, the redundancy in longitudinal CT sequence data, and the misalignment between model optimization goals and clinical priorities—which directly motivate and justify the study's core innovations, establishing a solid theoretical foundation with strong problem orientation that effectively bridges existing literature to the proposed research framework, making it an exemplary, purpose-driven scholarly review.

Originality and Independence.

This thesis demonstrates strong originality and independence: it innovatively proposes the DVT model combining Transformer architecture with secondary transfer learning to enable incidental thyroid cancer screening from routine chest CT, a previously underexplored clinical scenario; develops the CL-IJLO framework featuring a contrastive learning-based feature compression module that achieves 80% data reduction and a novel combined contrastive loss function that aligns model optimization with clinical priorities of high sensitivity; and establishes a unique three-dimensional research paradigm spanning spatial expansion, modal deepening, and temporal extension. Each core study maintains methodological independence with dedicated multi-center datasets, rigorous experimental designs, and separate validation protocols, while collectively forming a cohesive, original intellectual contribution that advances the concept of "one scan, multiple cancer screenings" beyond existing single-disease AI diagnostic approaches.

Arguments and Reasoning.

Arguments and Reasoning

This thesis constructs a rigorous, multi-layered argumentation framework following a deductive reasoning structure: it first establishes the clinical premise that lung and thyroid cancers impose severe disease burdens in China while conventional chest CT suffers from three fundamental limitations—single examination targets, underutilized information, and lack of dynamic assessment; then demonstrates through comprehensive literature review that advanced deep learning technologies including Transformers, contrastive learning, and transfer learning have demonstrated transformative capabilities in medical imaging but remain underexploited for addressing these specific bottlenecks; and subsequently validates three interconnected hypotheses through empirical evidence: (1) that spatial dimension expansion via the DVT model, combining Transformer architecture with secondary transfer

learning, enables accurate incidental thyroid cancer screening from routine chest CT (achieving 96% accuracy and 0.97 AUROC across multi-center datasets), (2) that temporal dimension extension via the CL-IJLO framework, featuring contrastive learning-based 80% feature compression and a novel combined contrastive loss function, achieves robust longitudinal pulmonary nodule malignancy prediction with 97% sensitivity and 2.1× training efficiency improvement, and (3) that modal dimension deepening through multimodal fusion offers a promising pathway for fine-grained lung cancer subtype classification; ultimately concluding that this unified "one scan, multiple screenings" paradigm represents a clinically viable, methodologically innovative approach to advancing early cancer detection, with each claim systematically supported by comparative experiments, ablation studies, and prospective independent external validation.

Evaluation within the Field.

This thesis represents a top-tier, paradigm-shifting contribution within the field of intelligent chest CT analysis, pioneering an innovative three-dimensional expansion framework spanning spatial, modal, and temporal dimensions that fundamentally redefines routine chest CT from a lung-only examination to a comprehensive multi-cancer screening platform; its significant methodological advancements include the novel application of contrastive learning for longitudinal CT sequence purification achieving 80% data compression, the combined contrastive loss function explicitly aligning model optimization with clinical priorities of high sensitivity, and the secondary transfer learning strategy effectively addressing the small sample high-noise challenge, all validated through rigorous multi-center prospective studies demonstrating state-of-the-art performance—96% accuracy and 0.97 AUROC for thyroid cancer screening and 97% sensitivity for pulmonary nodule malignancy prediction—while establishing a cost-effective, clinically viable approach to opportunistic cancer screening that delivers exceptional value particularly for resource-limited regions and provides reusable, generalizable technical principles that advance the broader medical imaging AI field.

Completeness and Coherence.

Completeness and Coherence

This thesis exemplifies exceptional completeness and coherence through its rigorously structured research design and tightly integrated logical architecture: it follows a complete academic thesis framework encompassing introduction, systematic literature review, three methodologically distinct yet thematically unified core research chapters, and comprehensive conclusion, with every component serving the overarching vision of advancing chest CT intelligence through the "spatial expansion → modal deepening → temporal extension" three-dimensional paradigm. The work demonstrates remarkable internal consistency as each chapter builds upon preceding arguments—the literature review precisely identifies the three critical research gaps that the subsequent chapters systematically address, while the unified theoretical foundation of deep learning innovation for clinical translation permeates all studies; methodologically, the thesis achieves full research closure for each investigation by consistently presenting clinical motivation, problem definition, dataset construction, model

design, experimental validation, comparative analysis, and interpretive discussion, ensuring no gaps in the reasoning chain from hypothesis formulation to empirical verification and clinical implication, ultimately delivering a cohesive, comprehensive body of work where individual technical contributions collectively converge toward the singular transformative goal of enabling "one scan, multiple cancer screenings."

Conclusion.

This is a high-quality doctoral thesis that has achieved an outstanding level for engineering and technology dissertations: it establishes a solid theoretical framework laying an excellent foundation for the entire research, demonstrates precise problem orientation by successfully justifying the necessity and innovation of the study, and adheres to sound academic standards with its complete structure, standardized citations, and self-consistent logic; the most prominent highlight and core feature lies in its exceptional "problem awareness"—rather than merely compiling literature, it systematically organizes existing research to naturally derive the three core innovations of this study, perfectly fulfilling the fundamental purpose of a literature review in connecting prior work to subsequent research and demonstrating the value of the investigation.

Additional Comments.